

Resilience of Networks in the B5G Era

Toward dynamic assessment of availability with Digital Twins

Presenter: Guillaume Fraysse

Orange Research, Châtillon, France

September 14th, 2026



Acknowledging the work of a whole team: current and past members

Current members from Orange team:

- Guillaume FRAYSSE, PhD
- Bertrand DECOCQ, PhD
- Benoit LEMOINE
- Firas MAALEJ, intern
- Christian TANGUY, PhD, HDR
- Ikram TEMERIGHT, PhD student

Past members:

- Rui LI, PhD
- Khaled SAYAD, PhD
- Manelle NOUAR



Acknowledging the work of a whole team: collaborations

Risk and Resilience of Complex Systems (RRSC)
chaire at CentraleSupélec:

- Anne BARROS
- Zhiguo ZENG
- Yiping FANG

Collaborative project:

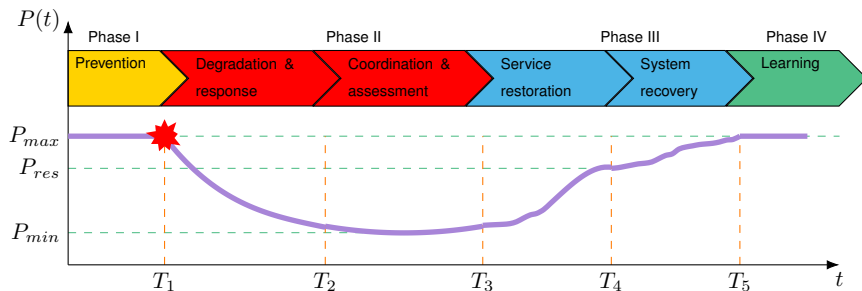
- SNS-JU Programmable Security Plane for 6G Ecosystems (MARE)



Definition

Resilience of networks (from ResiliNets project <https://resilinetts.org>)

Definition: *Resilience* is the ability of the network to face various incidents, maintain an acceptable level of service and return to normal operation.



From Nait Belaid et al [1]

How do we measure resilience?

Availability

A

Is the service operational ?

Uptime / downtime

Reliability

MTBF

How often does it fail?

MTBF / MTTF

Recoverability

MTTR

How fast can it be restored?

MTTR / TTR

Service continuity

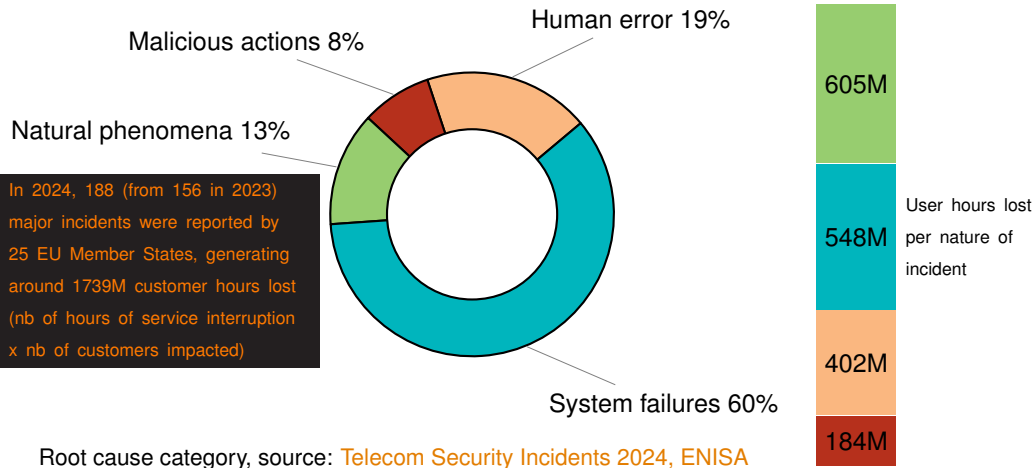
QoS

How much does service degrade?

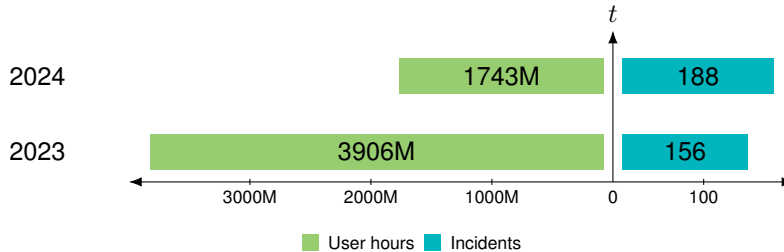
Latency / loss / throughput**Resilience**
assessment

$$A = \frac{\text{Uptime}}{\text{Observation time}}$$

Root Causes of incidents and their impact (2024)



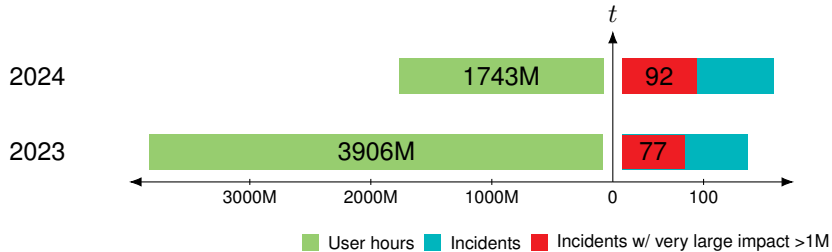
YoY evolution of incidents



Number of incidents submitted by countries and user hours lost (2023–2024)



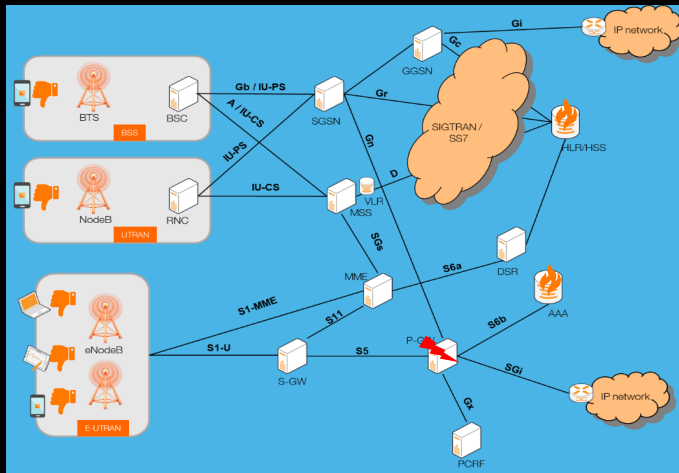
YoY evolution of incidents



Number of incidents submitted by countries and user hours lost (2023–2024)



Signaling storm : EE 4G network - March,2014



Redundancy and diversity were not sufficient anymore in 4G

Since 4G deployment, we face complex outages because of always-on devices



Redundancy and diversity were not sufficient anymore in 4G

Since 4G deployment, we face complex outages because of always-on devices

Trigger

- A failure in the network
- Devices are detached

Redundancy and diversity were not sufficient anymore in 4G

Since 4G deployment, we face complex outages because of always-on devices

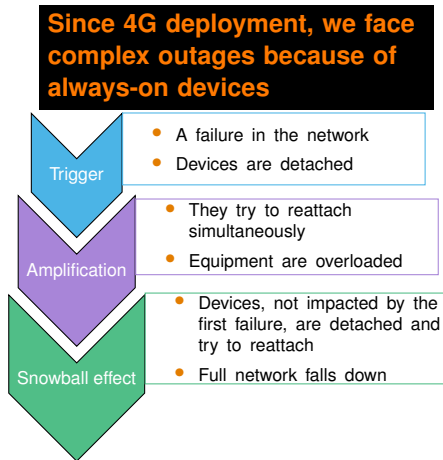
Trigger

- A failure in the network
- Devices are detached

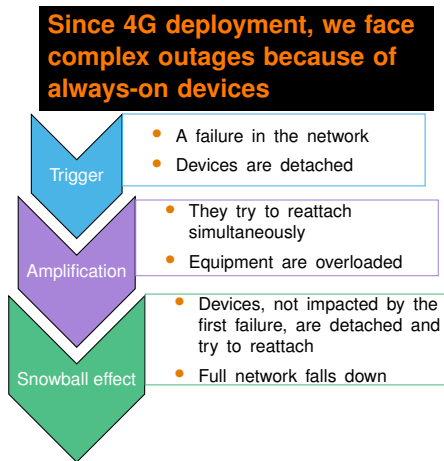
Amplification

- They try to reattach simultaneously
- Equipment are overloaded

Redundancy and diversity were not sufficient anymore in 4G



Redundancy and diversity were not sufficient anymore in 4G



4G standards improvement /
coordination mechanisms - timers

Industrial Implementation

Deployment

**Lessons learned for
5G standards design**

Resilience improvement for 5G

Redundancy/diversity

1

User Plan

New mechanisms allow a real user plan redundancy from an E2E perspective, combining access and core mechanisms

Redundancy/diversity

2

NF sets

NF (Network Function) can be organized as group of instances to allow load balancing, to react in case of failure of one instance

Load/Overload control

3

NF informs

Local or global mechanisms can be launched by an overloaded NF alerting its "neighbors" about its situation : timers, prioritization,...

4

Service continuity

Stateless NF

States information are stored in a separated database. In case of NF failure, another can take the floor retrieving the information from the database. No impact on the service

5

Dynamicity

Self healing...

Dynamicity is ensured by restoration mechanisms at 5G level and/or by NF instantiation on the fly at virtual machines or containers level

A 5G network is a system of systems

Network resilience can be affected by many parameters

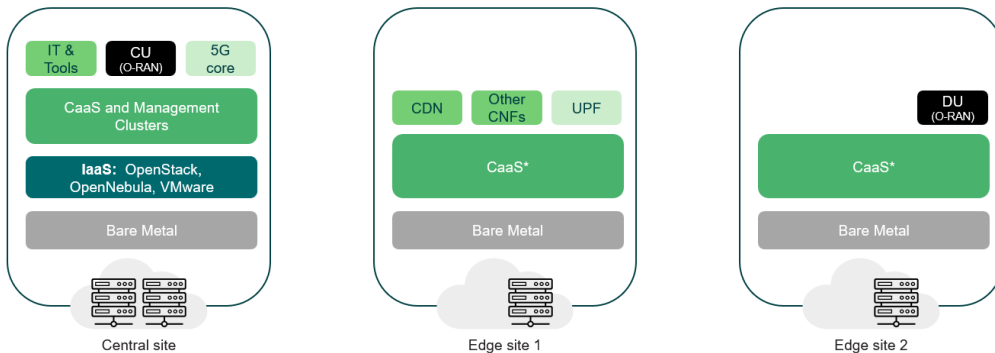
- hardware issues
- software issues (configuration, bug, ...)
- user traffic
- lifecycle management: updates, upgrades,
- weather conditions

5G and beyond networks are now built different

- Virtualization with Network Function Virtualization (NFV) and Software-Defined Networking (SDN)
- Slicing
- Cloud native microservices



New sources of problems join the old ones



from <https://sylvaproject.org/architecture-benefits/>



New sources of problems

New problems to address for resilience

- More layers
- More dependencies
- More dynamicity
- More heterogeneity



New sources of problems

New problems to address for resilience

- More layers
- More dependencies
- More dynamicity
- More heterogeneity

but... Telco world is not fast fashion

- End of copper lines in France planned for 2030
- 2G and 3G, ... sunset while deploying newer generations
- Hardware, Software life cycles / Technical debt



New sources of problems

New problems to address for resilience

- More layers
- More dependencies
- More dynamicity
- More heterogeneity

but... Telco world is not fast fashion

- End of copper lines in France planned for 2030
- 2G and 3G, ... sunset while deploying newer generations
- Hardware, Software life cycles / Technical debt



Agence nationale de la sécurité
des systèmes d'information



Inspection générale de
l'administration



Conseil général de l'économie



Commissariat aux communications
électroniques de défense



Inspection générale des
affaires sociales

RAPPORT

ÉVALUATION DE LA GESTION PAR L'OPÉRATEUR ORANGE
DE LA PANNE DU 2 JUIN ET DE SES CONSÉQUENCES
SUR L'ACCÈS AUX SERVICES D'URGENCE

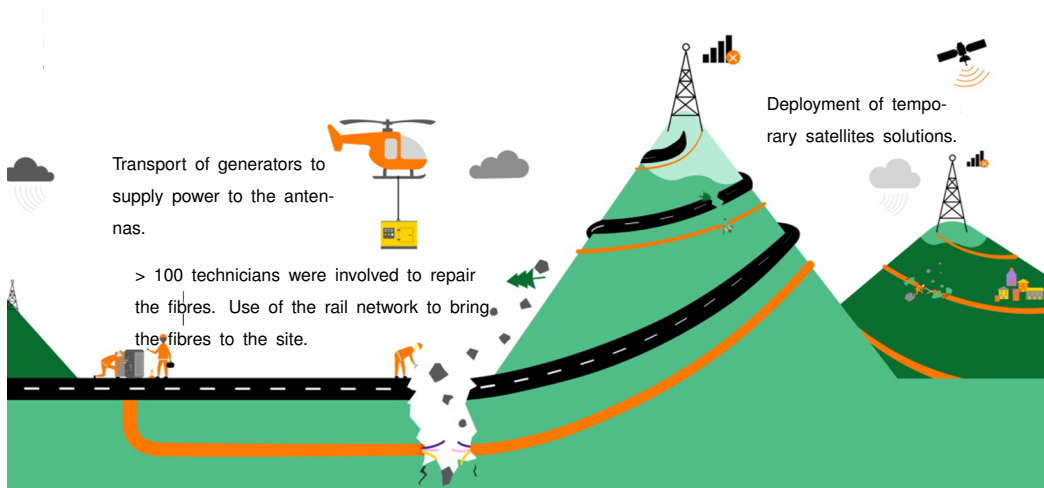
Paris, le 19 juillet 2021

51 boulevard de La Tour-Maubourg - 75006 PARIS 06 SP

Flood in la vallée de la Vésubie, October 2020



Flood in la vallée de la Vésubie, October 2020



New, and future, problems



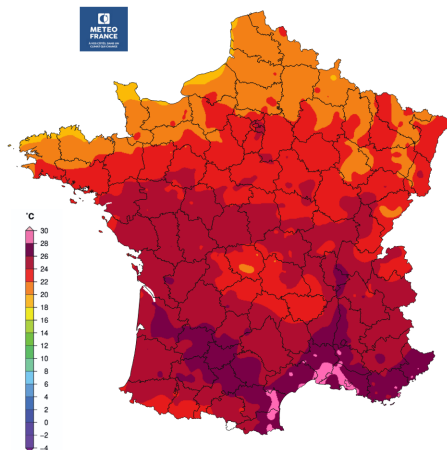
Mayotte, a French department in the Indian Ocean, was hit by tropical cyclone Chido on December 14th, 2024.

51 out of 54 mobile network antennas are out of service, mainly due to a lack of power. **99% of fixed-line Internet customers** are disconnected.

Increased complexity for the coordination of emergency services.



New, and future, problems



July 2026 is the hottest month ever recorded in France, average temperature was **24.9°C**. Climate change leads to higher average temperatures, longer heat waves, increased risks of fires, . . . Impact on material, infrastructure, people, ...

Global warming is not an hoax



PLANÈTE • RÉCHAUFFEMENT CLIMATIQUE

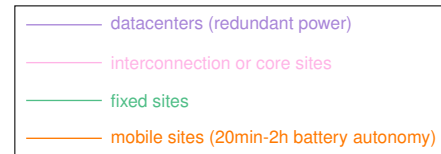
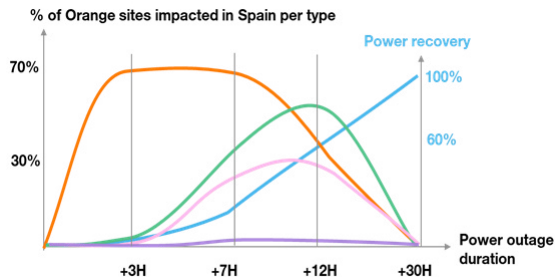
Août 2026 est le mois le plus chaud jamais mesuré dans le monde, Copernicus alerte sur des risques accrus pour l'Europe



Exogenous causes and consequences

Networks are interdependent

- Telecommunication networks require energy / Energy networks require transmission of information
- Transportation needed to access sites / Vehicles and infrastructure require connectivity



Impact of April 2025 power blackout in Spain on Orange sites, [Revue de l'Electricité et de l'Electronique 01/2026](#)

Modeling Formalisms: An End-to-End View

Non state-space models

- Fast and simple to solve
- Limited ability to capture dynamic behavior

Examples: RBDs, reliability graphs, fault trees

State-space models

- Capture state, time, and component dependencies
- May suffer from **state-space explosion**

Examples: CTMCs, Petri nets

Multi-level models

- Combine state-space and non state-space models
- Suitable for large and complex infrastructures

Also called hierarchical models

Key trade-off

Simple models are efficient but less expressive; dynamic models are more realistic but can become difficult to analyze as the number of states grows.

State of the Art (partial)

Availability modeling in 5G networks

Reference(s)	Model	Network part / service
Kaada et al., 2022, 2023 [2, 3]	CTMC	RAN
Temericht et al., 2025 [4]	RBD + CTMC	5G Service Function Chain
Soares et al., 2019[5]	RBD–SAN	Virtualized services / VNFs
Matos et al., 2019 [6]	RBD–FT–CTMC	IP Multimedia Subsystem
Khorsandroo et al., 2021 [7]	RBD–SRN	PDU session establishment
Dantas et al., 2021 [8]	FT–CTMC	Medical edge network
Matos et al., 2017 [9]	RG–FT–SRN	Cloud data-center network
Pathirana et al., 2024 [10]	FT–SAN	5G-MEC / O-RAN
Li et al., 2022, 2023 [11, 12]	Three-level TSCPN	SFC, VNF, and microservice layers

Main observations

- Strong focus on **SFCs**, VNFs, microservices, cloud, edge, and 5G Core services.
- Various dependability models are used: **CTMC**, **RBD**, **FT**, **SAN**, **SRN**, and Petri nets.
- Limited number of work on **RAN**.

Selected publications

- [2] S. Kaada et al. “Resilience Analysis and Quantification Method for 5G-Radio Access Networks”. In: *2022 IEEE Network of Future Conference (NoF)*. Oct. 2022, pp. 1–9.
- [3] S. Kaada et al. “Measuring 5G-RAN Resilience Using Coverage and Quality of Service Indicators”. In: *2023 IEEE/IFIP Network Operations and Management Symposium (NOMS)*. 2023, pp. 1–7.
- [4] I. Temericht et al. “Optimizing 5G Network Availability: Comparison Between Reliability Block Diagrams and Markov Chains Approaches”. In: *Proceedings of the 35th European Safety and Reliability Conference (ESREL 2025) and the 33rd Society for Risk Analysis Europe Conference (SRA-E 2025)*. June 15–19, 2025. Stavanger, Norway, June 2025.
- [5] C. Soares, J. Carapinha, and R. Aguiar. “Network-Aware Availability Modeling of an End-to-End NFV-Enabled Service”. In: *Proceedings of the IEEE International Conference on Dependable Systems and Networks (DSN)* (2019).



Selected publications (cont.)

- [6] R. Matos, N. Fonseca, and P. Maciel. “Availability Evaluation of a Virtualized IP Multimedia Subsystem for 5G Network Architectures”. In: *Proceedings of the IEEE International Conference on Communications (ICC)*. 2019, pp. 1–6.
- [7] S. Khorsandroo et al. “Performance and Availability Challenges in Designing Resilient 5G Architecture”. In: *IEEE Access* 9 (2021), pp. 1–15.
- [8] M. Dantas et al. “A Hierarchical Model Based on Availability and Resiliency Evaluation of Medical Edge Networks”. In: *Proceedings of the IEEE International Symposium on Network Computing and Applications (NCA)*. 2021, pp. 1–8.
- [9] R. Matos et al. “Reliability and Availability Evaluation for Cloud Data Center Networks Using Hierarchical Models”. In: *IEEE Systems Journal* 11.4 (Dec. 2017), pp. 2182–2193.
- [10] T. Pathirana, G. Nencioni, and R. F. Olimid. “Security-Aware Availability Modeling of a 5G-MEC System”. In: *arXiv preprint arXiv:2406.17554* (June 2024). DOI: [10.48550/arXiv.2406.17554](https://doi.org/10.48550/arXiv.2406.17554).



Selected publications (cont.)

- [11] R. Li et al. “A Petri Net-Based Model to Study the Impact of Traffic Changes on 5G Network Resilience”. In: *Proceedings of the 32nd European Safety and Reliability Conference (ESREL 2022)*. Dublin, Ireland, Aug. 2022, pp. 3016–3023.
- [12] R. Li et al. “Estimating 5G Network Service Resilience Against Short Timescale Traffic Variation”. In: *IEEE Transactions on Network and Service Management* (2023), pp. 1–11.



Aim

- Find exact or good estimates availability of the system (or service) availability (analytical expressions a plus)
- Identify the key parameters of the problem under consideration
- Go beyond the Markov approximation, which is not always realistic and may overestimate reliability
- Provide a better understanding that can be applied in ensuing simulations and policies (digital twins, maintenance)



Selected research activities



Meshed networks [34]

The exact expression of the end-to-end availability between S and T can be obtained, not by using series-parallel simplifications, but from a generic solution for recursive families of graphs (917 terms)

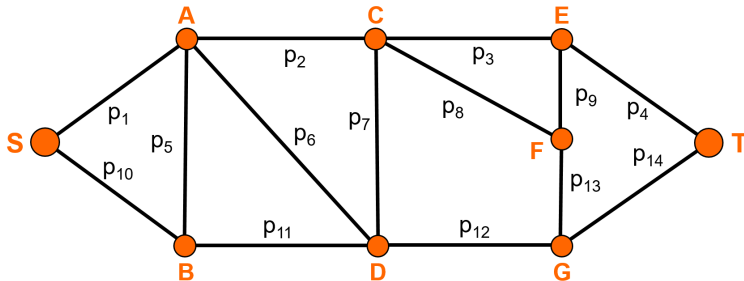


figure from [35]



Meshed networks [34]

The exact expression of the end-to-end availability between S and T can be obtained, not by using series-parallel simplifications, but from a generic solution for recursive families of graphs (917 terms)

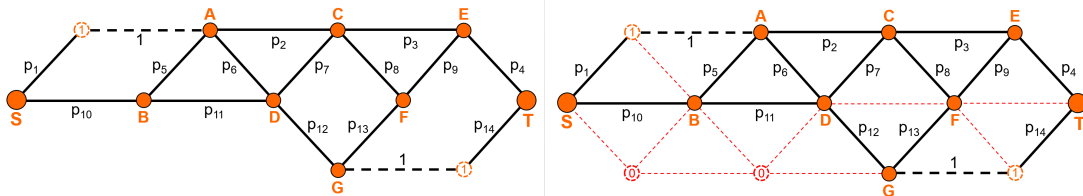
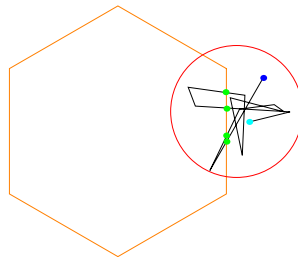
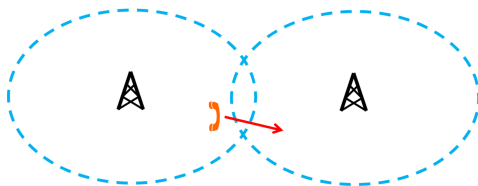


figure from [35]



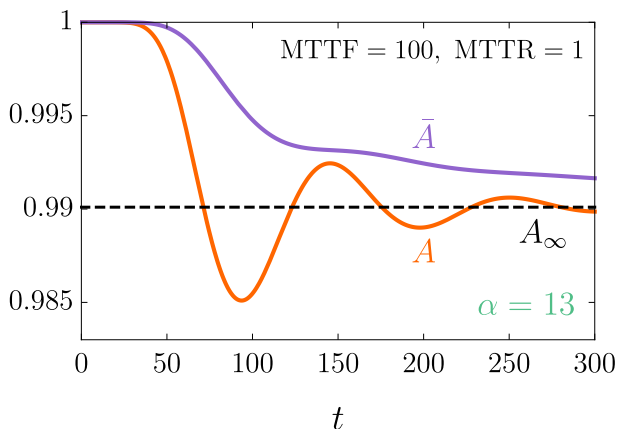
Hand-over rate and Random Waypoint model [31]



- What is the probability of leaving a cell?

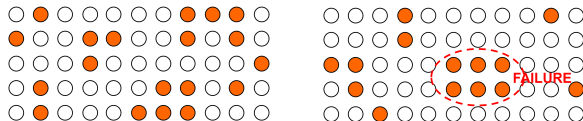


Importance of transients [29]

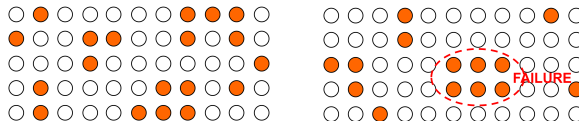


Instant and averaged availabilities $A(t)$ and $\bar{A}(t)$ for a gamma($\lambda, 13$) lifetime distribution and an exponential repair distribution

Two-dimensional (3,2)-out-of-(11,5):F system [30]



Two-dimensional (3,2)-out-of-(11,5):F system [30]



$$\begin{aligned}
 \mathcal{R} = & (1 - q)^6 \left(1 + 6q + 21q^2 + 56q^3 + 126q^4 + 252q^5 + 426q^6 + 576q^7 \right. \\
 & + 563q^8 + 205q^9 - 651q^{10} - 2111q^{11} - 3865q^{12} - 4912q^{13} - 3949q^{14} \\
 & - 111q^{15} + 5867q^{16} + 12046q^{17} + 15332q^{18} + 12047q^{19} + 1086q^{20} \\
 & - 14387q^{21} - 25509q^{22} - 24554q^{23} - 11602q^{24} + 7156q^{25} + 22184q^{26} \\
 & + 27253q^{27} + 18578q^{28} - 240q^{29} - 16433q^{30} - 19156q^{31} - 9994q^{32} \\
 & + 856q^{33} + 6078q^{34} + 5867q^{35} + 3723q^{36} + 731q^{37} - 1794q^{38} \\
 & - 1925q^{39} - 344q^{40} + 406q^{41} + 69q^{42} - 38q^{43} + 21q^{44} \\
 & \left. + 41q^{45} + 12q^{46} - 4q^{47} - 4q^{48} - q^{49} \right)
 \end{aligned}$$

where q is the unavailability of each element



Selected publications

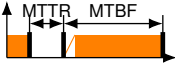
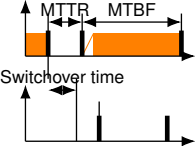
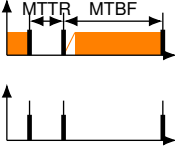
- [28] Christian Tanguy. “When Considering the Asymptotic Availability Is Not a Safe Bet”. In: *Proceedings of the 30th European Safety and Reliability Conference and the 15th Probabilistic Safety Assessment and Management Conference*. (Venice, Italy). Research publishing, singapore, 2020, pp. 3091–3098. ISBN: 978-981-14-8593-0.
- [29] Christian Tanguy. “Transient Behaviour of Instantaneous and Average Availabilities in Non-Markovian Configuration”. In: *Safety and Reliability of Systems and Processes* (2022), pp. 185–194. DOI: [10.26408/srsp-2022-13](https://doi.org/10.26408/srsp-2022-13).
- [30] Christian Tanguy. “Impact of Common-Cause Failures on the Availability of Connected (r, s)-out-of-(m, n): F Systems”. In: *35th European Safety and Reliability Conference (ESREL 2025) and the 33rd Society for Risk Analysis Europe Conference (SRA-E 2025)*. 35th European Safety and Reliability Conference (ESREL 2025) (Stavanger, Norway). RESEARCH PUBLISHING, SINGAPORE, 2025. ISBN: 978-981-94-3281-3. DOI: [10.3850/978-981-94-3281-3-proceedings](https://doi.org/10.3850/978-981-94-3281-3-proceedings).

Selected publications (cont.)

- [31] Christian Tanguy. “Handover Rate in Wireless Networks and Random Waypoint Mobility Model: An Almost Simple Result”. In: 10th International Conference on Mathematical Methods in Reliability (Grenoble, France). 2017.
- [32] Christian Tanguy. “Shannon Entropy and Its Variants for the Maintenance of Large Meshed System Architectures”. In: *Proceedings of the 13th IMA International Conference on Modelling in Industrial Maintenance and Reliability 2025*. 13th IMA International Conference on Modelling in Industrial Maintenance and Reliability (MIMAR) (Nancy, France). Institute of Mathematics and its Applications, 2025. DOI: [10.19124/ima.2025.01](https://doi.org/10.19124/ima.2025.01).
- [34] Christian Tanguy. “What is the probability of connecting two points?” In: *Journal of Physics A: Mathematical and Theoretical* 40.47 (2007), pp. 14099–14116.

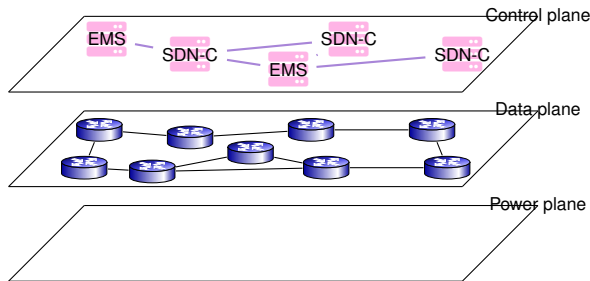


Typical resilience mechanisms for telcos

		Scheme	Backup ratio	Failure-Repair Sequence	Interruption Time
	App. Server	Self Healing	0		MTTR
	5G User Plane	N:B Shared Protection	B/N		switchover time
	5G Control Plane	1+1 active/active protection	1		0

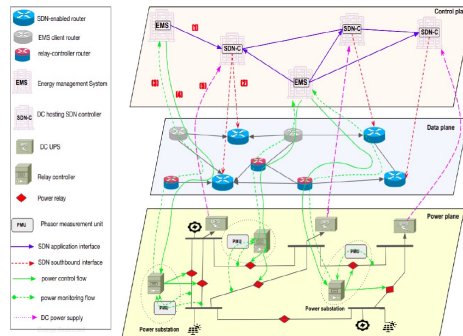
Cross-domain resilience in Cloud-native, Critical Cyber Physical Systems Networks

Results: Model for the propagation of failures between **interdependent** critical infrastructures and analyze its impact on overall availability. Optimize the placement of virtualized functions across multiple sites based on availability and latency constraints. Propose an architecture for exchanging availability information to plan maintenance operations.



Supervised Learning of virtualization infrastructure availability

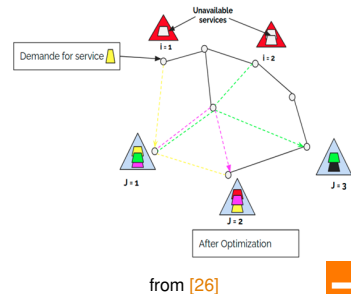
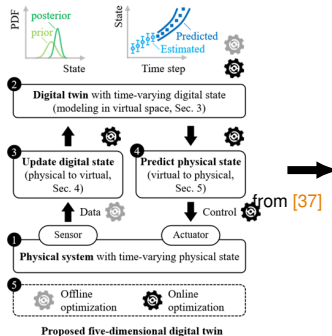
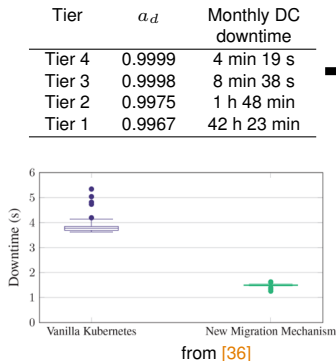
<https://theses.hal.science/tel-04713044>



From Khaled Sayad et al [27]

Supervised Learning of virtualization infrastructure availability

Objective: Model the failure statistics of infrastructure components and develop a supervised learning approach for the model parameters-aimed at compressing measurements from infrastructure sensors for a Digital Twin-while retaining only information regarding rare events such as failures, resilience, and the return to normal operations.



Selected publications

- [13] Benoît Lemoine. “Energy Efficiency of N:1 Protection Setups with Kubernetes Horizontal Pod Autoscaler”. In: *2022 25th Conference on Innovation in Clouds, Internet and Networks (ICIN)*. 2022 25th Conference on Innovation in Clouds, Internet and Networks (ICIN). Mar. 2022, pp. 126–130. DOI: [10.1109/ICIN53892.2022.9758129](https://doi.org/10.1109/ICIN53892.2022.9758129).
- [14] Benoît Lemoine. “Kubernetes Horizontal Pod Autoscaler Enhanced Design for Protected Workloads”. In: *2024 IEEE Symposium on Computers and Communications (ISCC)*. 2024 IEEE Symposium on Computers and Communications (ISCC). June 2024, pp. 1–7. DOI: [10.1109/ISCC61673.2024.10733692](https://doi.org/10.1109/ISCC61673.2024.10733692).
- [24] Khaled Sayad et al. *Availability Modeling and Analysis of Cloud-Native, Interdependent Cyber-Physical*. Feb. 16, 2024. DOI: [10.2139/ssrn.4728817](https://doi.org/10.2139/ssrn.4728817). Social Science Research Network: [4728817](https://ssrn.com/abstract=4728817). Pre-published.



Selected publications (cont.)

- [25] Khaled Sayad. “Cross-Domain Resilience in Cloud-native, Critical Cyber-Physical Systems Networks : Availability Modeling, Analysis, and Optimization of Critical Services Provisioning”. These de doctorat. université Paris-Saclay, Apr. 10, 2024. URL: <https://theses.fr/2024UPAST028> (visited on 07/17/2026).
- [26] Khaled Sayad et al. “Interdependency-Aware Resource Allocation for High Availability of 5G-enabled Critical Infrastructures Services”. In: *32nd European Safety and Reliability Conference, ESREL 2022*. Dublin, Ireland, Aug. 2022. DOI: [10.3850/978-981-18-5183-4_S28-03-640-cd](https://doi.org/10.3850/978-981-18-5183-4_S28-03-640-cd).
- [27] Khaled Sayad and Benoît Lemoine. “Towards Cross-domain Resilience in SDN-enabled Smart Power Grids: Enabling Information Sharing through Dataspaces”. In: *2023 IEEE International Conference on Omni-layer Intelligent Systems (COINS)*. 2023 IEEE International Conference on Omni-layer Intelligent Systems (COINS). July 2023, pp. 1–6. DOI: [10.1109/COINS57856.2023.10189319](https://doi.org/10.1109/COINS57856.2023.10189319).



Selected publications (cont.)

- [36] Davide Tazzioli, Riccardo Venanzi, and Luca Foschini. “Stateful service migration support for kubernetes-based orchestration in industry 4.0”. In: *2024 IEEE Symposium on Computers and Communications (ISCC)*. IEEE. 2024, pp. 1–6.
- [37] Adam Thelen et al. “A comprehensive review of digital twin—part 1: modeling and twinning enabling technologies”. In: *Structural and Multidisciplinary Optimization* 65.12 (2022), p. 354.
- [38] Ilyswarya Narayanan, Aman Kansal, and Anand Sivasubramaniam. “Right-sizing geo-distributed data centers for availability and latency”. In: *2017 IEEE 37th International Conference on Distributed Computing Systems (ICDCS)*. IEEE. 2017, pp. 230–240.



From availability assessment to automation

Availability can be evaluated using several **well-established approaches**.

Analytical
models

Markov
models

RBD / Fault Trees
reliability models

Experimental
evaluation

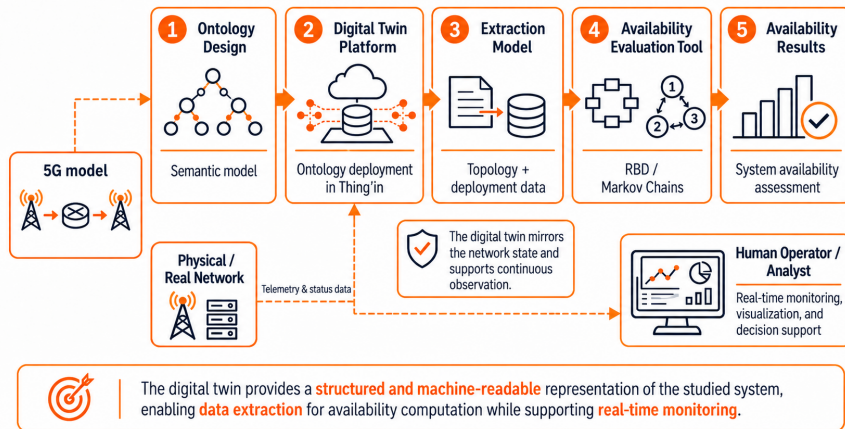
Practical challenge

The methods themselves are mature, but building a complete end-to-end assessment often requires manual effort for **model construction • scenario configuration • failure injection • data collection • analysis**.



Digital Twins provide a promising foundation for making availability assessment **systematic, repeatable and automatable**.

Digital twin technology



Ontology Design for the 5G Digital Twin

The ontology provides a **semantic representation** of the main entities, relationships and attributes of the virtualized 5G system.

Ontology classes

VNF ● Microservice ● Replica ● Pod ● Server

Object properties

hasMicroservice: VNF → Microservice

hasReplica: Microservice → Replica

deployedIn: Replica → Pod

hostedOn: Pod → Server

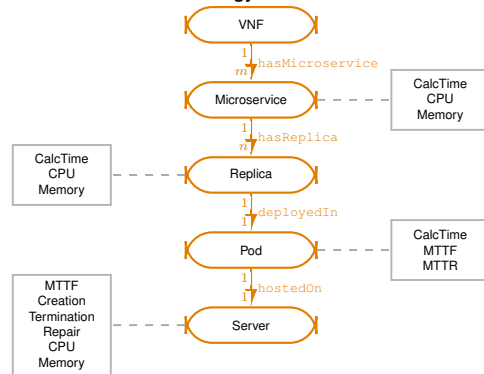
Data properties

Microservice / Replica: CalcTime, CPU, Memory

Pod: MTTF, MTTR, CalcTime

Server: MTTF, MTTR, Creation, Termination, CPU, Memory

Ontology schema



The ontology captures both the **logical service structure** and its **deployment infrastructure**.

From Digital Twin Data to Availability Evaluation

Extracted topology & redundancy

```
VNF: CU
  MicroserviceCU1: 2 replicas
  MicroserviceCU2: 2 replicas
  MicroserviceCU3: 2 replicas
VNF: DU
  MicroserviceDU1: 2 replicas
  MicroserviceDU2: 2 replicas
  MicroserviceDU3: 2 replicas
VNF: UPF
  Microservice1: 2 replicas
  Microservice2: 2 replicas
  Microservice3: 2 replicas
```

Extracted dependability parameters

Availability input parameters

```
Server MTTF      : 5000
Server repairTime: 1
Pod MTTF         : 200
Pod repairTime   : 0.00278
```

Failure and repair rates

```
lambda_server = 2.000000e-04 /h
mu_server     = 1.000000e+00 /h
lambda_pod    = 5.000000e-03 /h
mu_pod        = 3.597122e+02 /h
```

Inputs to the availability model

Availability Evaluation
RBD / Markov Chains

System Availability

99.999917%

Use case: TMForum : Living Network – Phase III Catalyst



The studied use case is composed of two service chains: a **User Plane Chain** and a **Control Plane Chain**.

User Plane Chain



Control Plane Chain



Modeling assumptions:
each **NF/VNF** is decomposed into **3 microservices**,
and each microservice is instantiated with **2 replicas**.

- [33] Ikram Temericht et al. “Optimizing 5G Network Availability: Comparison Between Reliability Block Dia-grams and Markov Chains Approaches”. In: (2025). URL: <https://rpsonline.com.sg/proceedings/esrel-sra-e2025/pdf/ESREL-SRA-E2025-P5793.pdf>.



Other actions attached to a chair

Risk and Resilience of Complex Systems (RRSC) Chair of CentraleSupélec, led by Prof. Anne Barros https://r3.centralesupélec.fr/en/Chair_RRCS



Scientific Project 2024-2029:

- Axis 1: System Modeling for Risk, Resilience, and Reliability
- Axis 2: Decision Support under Uncertainty
- Axis 3: Environmental and Societal Challenges



Partners



谢谢

Thank you

Merci

*"The greatest glory
of living lies not
in never falling but
in rising every time
you fall."*

Nelson Mandela

